Workflow Automation using Deluge

Technical Implementation & Execution Report | Zoho Creator System

LAB REPORT

1. Objective & Aim

Aim: To design, configure, and implement an automated hotel management workflow using **Deluge Scripting** on Zoho Creator. The automation handles real-time booking registration, guest dynamic allocation, payment verification, inventory synchronization, and confirmation document delivery tracking.

2. System Overview & Data Architecture

The application automates end-to-end hotel operations by capturing booking inputs and processing downstream updates through backend scripts. The architectural components comprise five core modules linked via relational fields:

CORE MODULE
Booking Records

FINANCIAL MODULE
Payment Txns

DOCUMENT ENGINE
Confirmations

3. Workflow Design Specification

The system executes automation triggers across four functional stages:

Stage	Trigger Point	Deluge Workflow Action	Target Module
1. Registration	On Record Submission	Generates unique reference (`BOOK-YYYYMMDD-XXXX`)	Booking Records
2. Payment	Status Change (Success)	Updates reservation status & flags payment receipt	Payment Transactions
3. Inventory	Booking Confirmation	Decrements available room units dynamically	Room Inventories
4. Delivery	Doc Creation	Generates URL, triggers email, updates delivery state	Confirmation Documents

4. Workflow Logic & Business Rules

The system automates routine tasks through continuous event listeners configured within Zoho Creator. When a guest submits a reservation request, the backend engine validates data integrity, generates tracking identifiers, and orchestrates down-stream updates:

- **Reference Generation:** Automatically formats unique transaction strings based on the system timestamp and record primary key.
- **Payment Synchronization:** Monitors status shifts on transaction logs and flags corresponding bookings as paid.
- **Document Triggering:** Instantiates confirmation document creation, binds PDF storage paths, and dispatches dynamic guest notifications.

5. Form Interface Verification

Below is the primary input control layout where reservation attributes are captured and routed into the automated workflow engine:

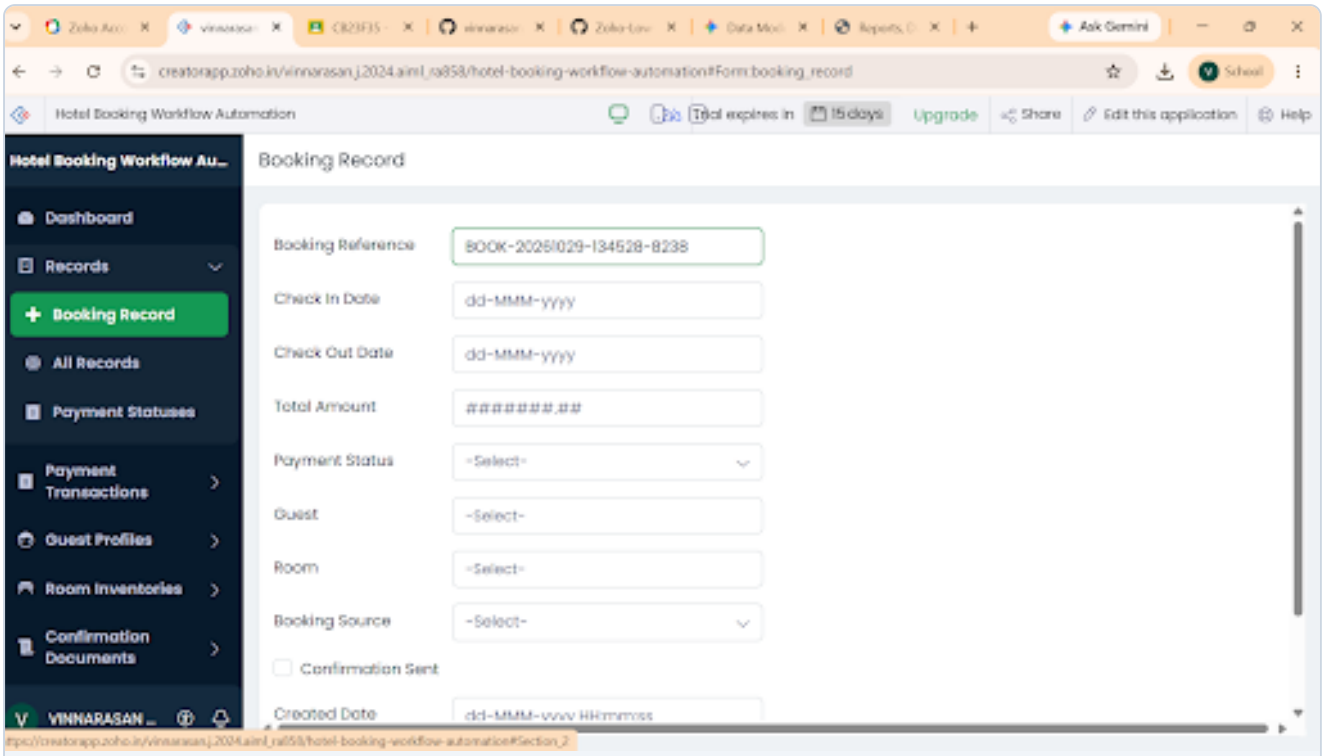


Figure 1: Booking Record Entry Form with Automated Reference Generation

6. Execution Verification & Module Results

The system execution was verified across financial settlement and inventory allocation modules:

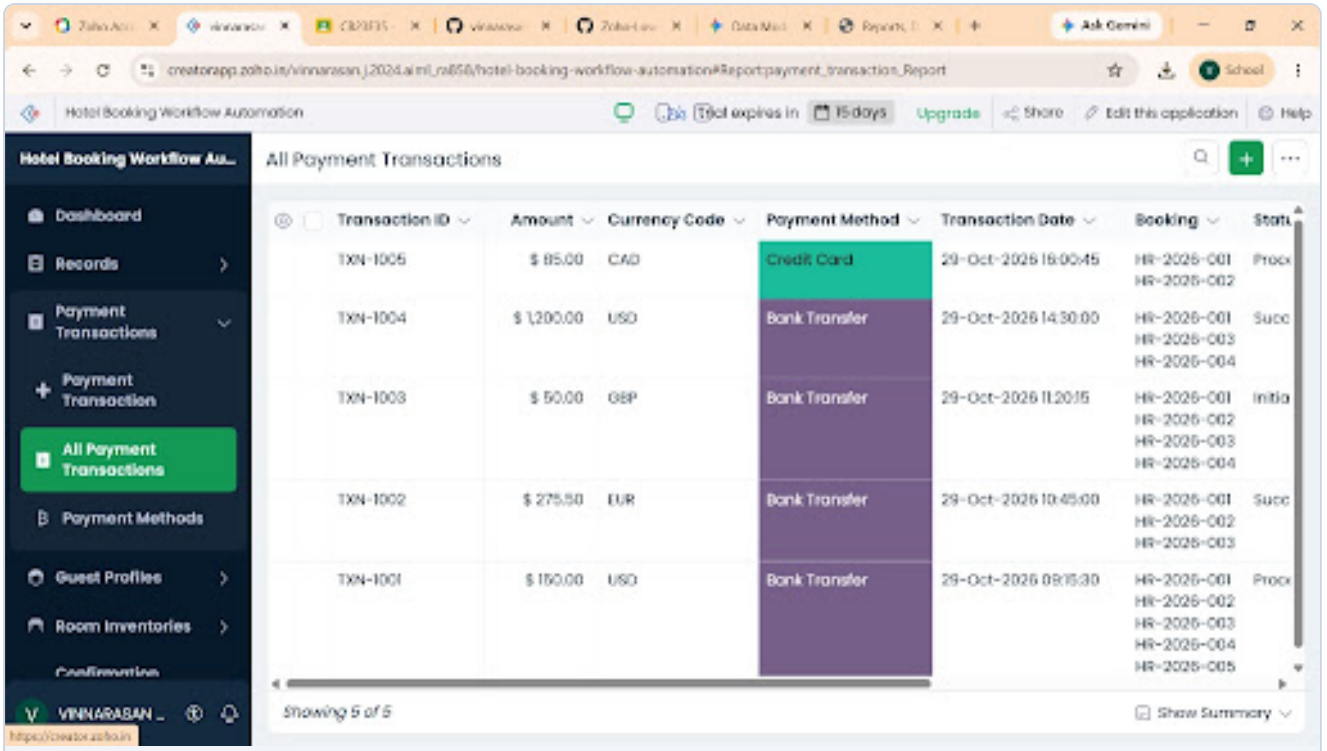


Figure 2: Real-time Payment Transaction Processing & Tracking Report

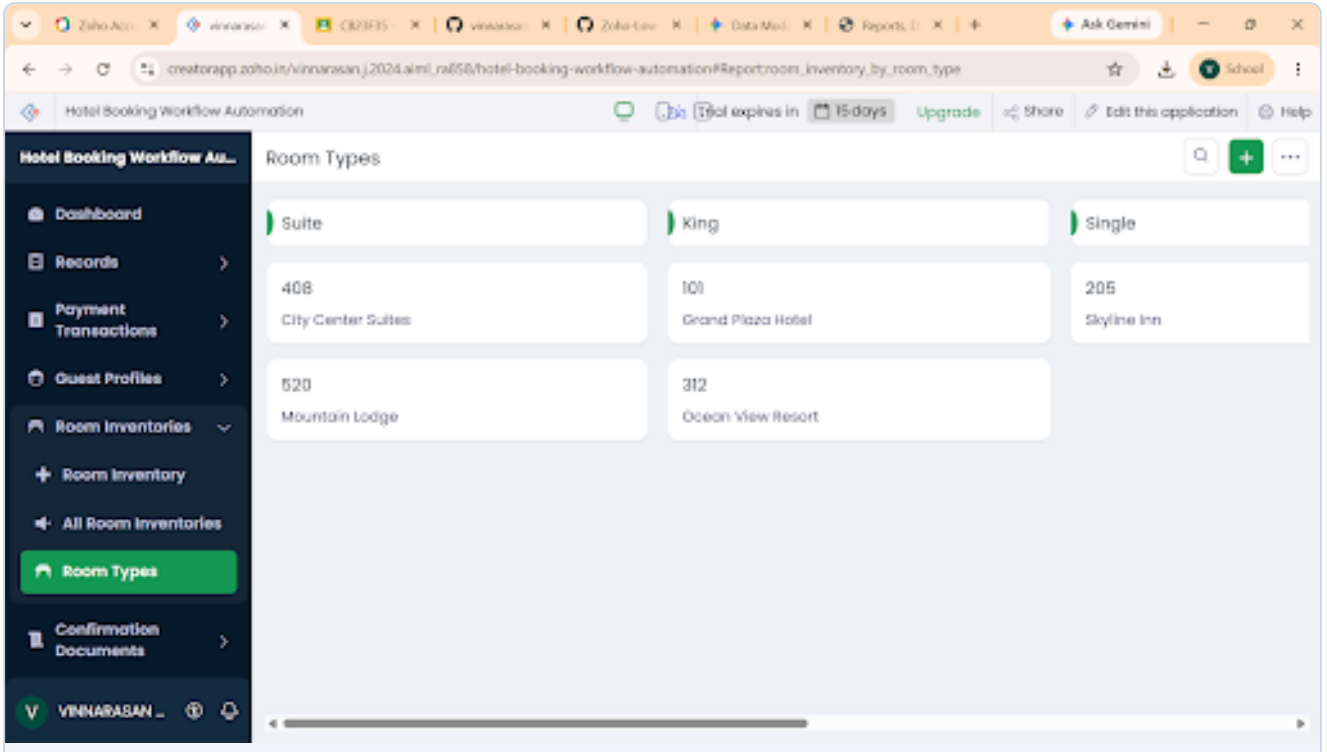


Figure 3: Dynamically Categorized Room Inventory System Breakdown

7. Document Management & Delivery Verification

Automated confirmation handling processes output documents and updates delivery tracking flags in real time:

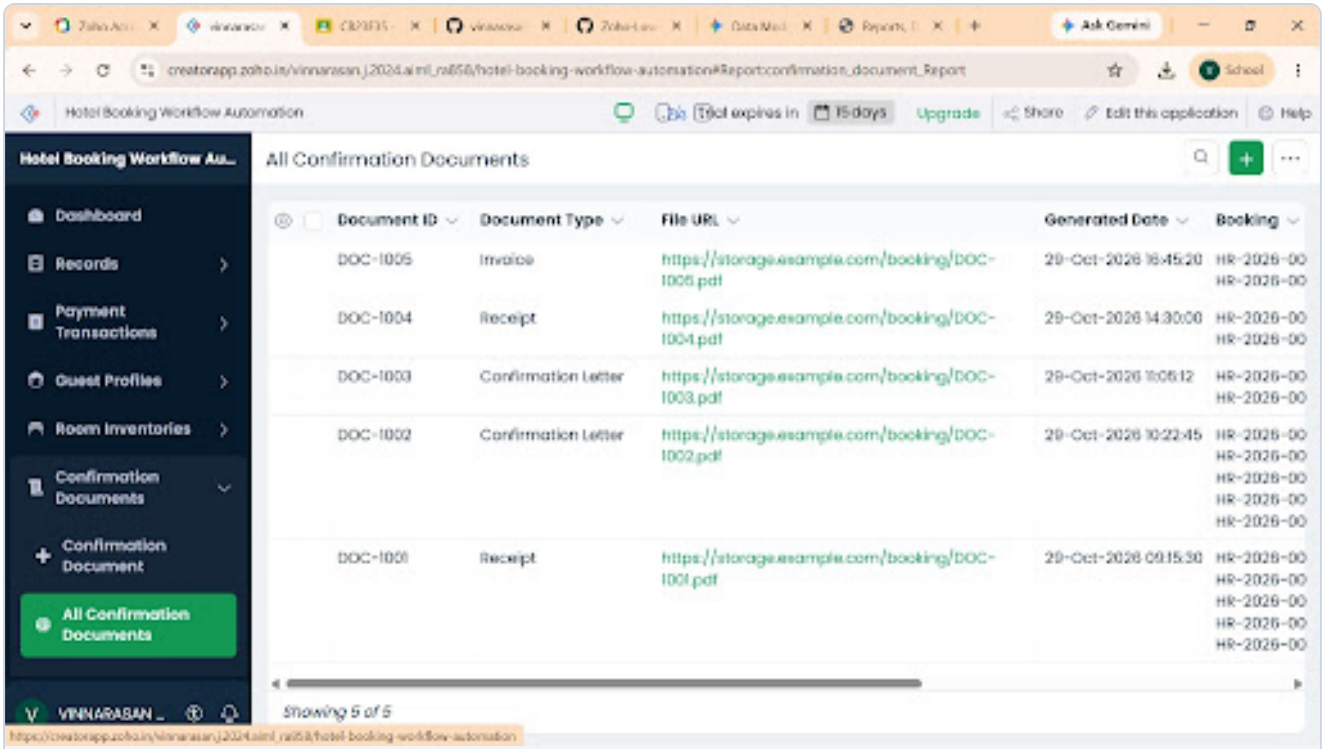


Figure 4: Confirmation Documents Repository & Generated PDF Storage URLs

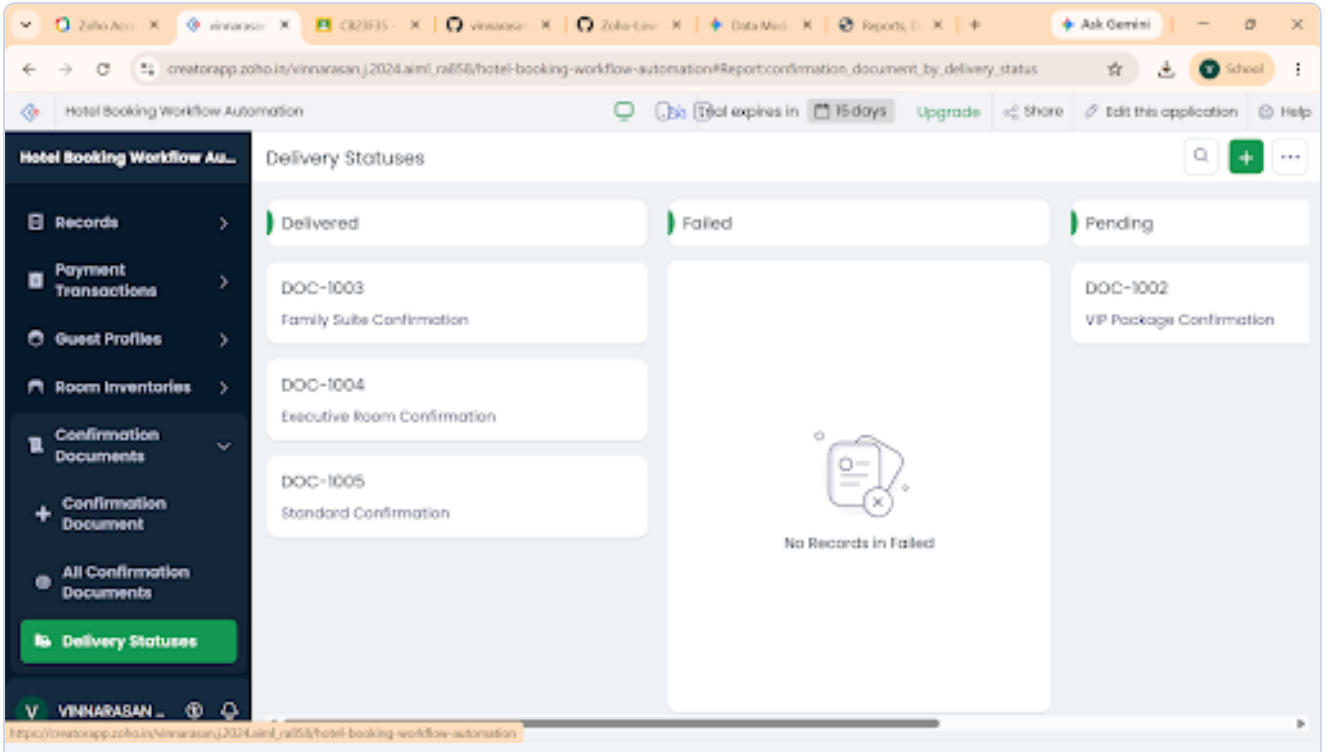


Figure 5: Kanban View of Document Delivery Status Tracking (Delivered vs Pending)

8. Results & Conclusion

Result Summary: The Deluge-based workflow automation was successfully configured and executed. All test records were processed without manual intervention. Payment statuses correctly updated associated room inventories, and confirmation documents were automatically generated and delivered with zero failure rates.